

A Small, Low-Cost First Test of an Idea to Help Rats Learn

Introduction in simple words · Researcher: Solano Moreira Furlan

What is the idea?

Scientists found that a natural substance in the fluid around young brains, called FGF17, seems to help old mice remember things better. Another substance, called 7,8-DHF, is thought to help brain cells make stronger connections with each other. We want to know: if we give both, and also give the animals a rich, stimulating place to learn, do they become better at solving different kinds of problems than normal rats of the same age?

What will we do?

This is a deliberately small version. We use 30 healthy young rats in three groups of 10: normal rats; rats that spend time in the learning environment; and rats that get the two substances and the learning environment. Another 18 rats are helpers: 12 are used to check that FGF17 really does something in the rat brain, and 6 for practice and safety checks. All 30 main rats do the same three tasks — a puzzle where the rules suddenly change, a maze with new problems, and a swimming-pool memory test. Two halves of the group take the tasks in a different order, at the same time, to save weeks. The check on FGF17 comes first: if there is no sign that it works, we stop and save the money. Some scientists doubt that 7,8-DHF works the way people first thought, so we treat that as an open question. The rats get a small operation that places a thin tube into the fluid space of the brain, daily injections for five weeks, limited food for several weeks and swimming tests, and are humanely put to sleep at the end; an animal-ethics committee must approve everything first.

Who does the work?

A good university laboratory in Brazil will run the whole experiment, using this document together with detailed lab procedures that the lab staff write and the researcher approves. The researcher guides it from a distance and keeps the secret group codes, so the staff giving the medicines do not know which rat got what (they can see which rats live in the enriched space), and the researcher checks the results independently.

How much will it cost?

Materials only (animals, chemicals, supplies) are planned at about USD 7,300. It could be as low as 6,000 or as high as 8,800, and up to about 10,300 if an import tax exemption is refused; all of these are below the researcher's ceiling of USD 12,000 for materials. Staff pay and the university's running-cost charges are NOT included. We roughly expect them to add USD 8,000 to 20,000 (an unverified guess), so the real total might be about USD 15,000 to 30,000. The lab must quote them.

What will this test tell us — and not tell us?

It can tell us whether FGF17 shows a chemical sign that it is working in the rat brain, whether the lab can run all the tasks smoothly, and whether the results lean in a promising direction. With so few rats, only a very large effect would show up clearly, and we cannot tell the effect of the substances apart from the effect of the environment. It cannot prove that the idea works, and it cannot say anything about people.

What if it goes well?

Then we plan a bigger, stronger experiment. Its first part would separate what each ingredient does. Its second part would use optogenetics — a new technique that switches chosen brain cells on and off with light — to see how brain activity interacts with these substances.

The long-term dream

In the far future we hope this kind of approach might help people who have learning difficulties or problems with how brain cells connect, and older people with Alzheimer's disease, by helping the brain rebuild connections. That is a hope, not a result. Many steps, long safety testing, and strict ethics rules come first, and there are real obstacles (for example, how to get these substances into a human brain safely). The rest of this document gives the technical details.

Technical protocol (Brazil mini proof of concept; core consumables ≈ USD 7.3k, range 6.0–8.8k)

Does a candidate plasticity-promoting treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive tasks in rats? A minimal, staged proof-of-concept study run by a Brazilian host laboratory under remote oversight.

Researcher / sponsor	Solano Moreira Furlan (remote; the host laboratory runs the experiment)
Document	Brazil mini proof-of-concept protocol, version 1. The task methods, statistics logic and many safeguards come from a larger design reviewed in eight rounds in its India form and one round in its Brazilian full-size form; this mini design is reviewed in the rounds listed in section 11.
Date	9 October 2026
Status	Not approved by an ethics committee (CEUA). Brazil-specific rules and prices come from search snippets and reviewer knowledge and must be confirmed locally. Prices are estimates, not quotes. “crf” in the request was read as CSF (assumption to confirm).

Executive summary (technical)

Question. Does a candidate plasticity-promoting treatment — FGF17 (identified in young cerebrospinal fluid; Iram 2022 [1]) plus 7,8-DHF (a putative TrkB/BDNF-pathway agonist whose mechanism is disputed [2–4]) — combined with a structured enriched learning environment (ELE) raise the performance of healthy young rats across three tasks, one of which (set-shifting) uses stimulus dimensions not practised in the ELE, compared with untreated peers?

Design. One wave. Three arms of 10 enrolled rats (30 main rats; 6–7 expected completers per arm): C (vehicle, standard housing), E (vehicle + ELE) and D+E (FGF17 + 7,8-DHF + ELE). There is no drug-only arm, so drug and environment cannot be separated; the study compares the combination with enrichment alone and with untreated peers. 48 rats in total (30 main, 12 FGF17-engagement satellites, 6 practice/tolerability), plus 12 contingency rats registered as a maximum (60 at most). Phase 0a (reagent activity and engagement; a gate) precedes Phase 0b (behaviour). The two halves of the cohort (15 rats each, balanced by arm) run the three tasks in parallel and in different orders, which shortens the calendar to about 8 weeks. The study informs, and does not confirm, whether a larger study is justified.

Budget (honest). Consumables-only core ≈ USD 7.3k base; ≈ 6.0k optimistic; ≈ 8.8k pessimistic; up to ≈ 10.3k if the CNPq import exemption is refused. All are inside the researcher-approved USD 12k materials ceiling. Personnel, service fees and institutional overhead are not included and must be quoted by the host; the all-in cost is probably USD 15–30k (unverified).

1. Aim, hypotheses and what the data can claim

Primary aim (feasibility): show that the reagent is active and engages its pathway in vivo; show that a Brazilian host can run the ELE and all three tasks with an acceptable completion rate; estimate variance and the direction of the D+E – C contrast.

Exploratory hypothesis: D+E performs better than control peers (C) and than enrichment alone (E).

Transfer: set-shifting is the primary transfer readout (odour and digging-medium dimensions are excluded from the ELE). Hebb-Williams and the water maze are spatial; the ELE has tunnels, climbing and wheels, so they are flagged as spatial, not transfer.

Allowed claims: feasibility, attrition, variance, direction with a 95% interval of D+E – C and D+E – E (overall index; set-shifting reported separately), and pathway engagement (pERK at the ventricular interface) — for the drug package as a whole, in combination with the ELE.

Not allowed: “smarter”, “general intelligence” or a g factor [8,9]; neuroplasticity; single-agent claims; any separation of drug effect from environment effect; TrkB-specific claims (pERK shows pathway engagement only); human or disease claims; replication of Iram 2022 (an aged-mouse rescue; this asks about enhancement in healthy young adults, which no study found has tested).

2. Long-term goal, next experiment (including optogenetics) and translational roadmap

Recommended wording (translational reviewer, adopted):

“Our long-term aim is to test whether combining a putatively plasticity-promoting biological treatment (an FGF17-type factor plus a validated TrkB/BDNF-pathway agent) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer’s disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Prior neurotrophin programmes in Alzheimer’s have not shown benefit (including a direct-delivery AAV-NGF trial), brain delivery of protein factors is a major obstacle, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates.”

Stage	Content	Gate to proceed
0 – this study	0a engagement/reagents → 0b behaviour in healthy young rats (mini proof of concept; tissue banked, histology not included).	Section 8.
1a – next experiment: powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment (and exercise); sustained infusion by pump; dose-response; durable vs on-drug effects; TrkB and FGFR necessity tests; synaptic readouts (PSD-95/synaptophysin on banked tissue, LTP).	Replicated effect on a pre-registered primary; structural effect.
1b – next experiment: optogenetics (only if 1a is positive; mechanistic, not translational)	Question: how does controlled neuronal activity interact with FGF17 and with 7,8-DHF-driven plasticity? Candidate designs (to be designed and reviewed in their own round): (a) activity × plasticity-opening — optogenetic stimulation of a defined circuit (e.g. CaMKIIa-driven channelrhodopsin in prefrontal or hippocampal pyramidal neurons) during or after learning, with and without FGF17/7,8-DHF, asking whether activity amplifies oligodendrocyte-precursor proliferation, myelination and synaptic plasticity (mouse precedent: premotor-cortex stimulation, Gibson 2014 [10]; extension to prefrontal/hippocampal circuits is an extrapolation); (b) necessity — inhibitory opsins during learning or consolidation; (c) cell-type targeting of oligodendrocyte-lineage cells needs Cre lines that are mostly mouse (species switch or an AAV-based rat strategy). Requirements: AAV vectors with institutional biosafety approval (CIBio/CQB; CTNBio authorisation for the higher risk classes under Lei 11.105/2005; the facility’s CQB must cover genetically modified animals at the right containment level), fibre implants, calibrated light sources, opsin-negative/fluorophore-only and light-only controls (heating, visual artefacts), histological verification of opsin and fibre placement, and monitoring for stimulation-induced seizures/kindling. Beyond this budget; own funding, equipment, expertise and ethics approval.	1a positive; opsin tools validated in the host lab.
2 – disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer’s models (5xFAD, APP/PS1, TgF344-AD) — amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models at appropriate ages: Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded) as separate tracks.	Benefit with engagement and structural evidence.

Stage	Content	Gate to proceed
3 – drug development	Clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — unproven); a validated TrkB-pathway drug with human-relevant PK; two-species toxicology; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (ANVISA clinical-development dossier, DDCM under RDC 945/2024; ethics review within the national human-research ethics system, Lei 14.874/2024; US IND / EU CTA).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, cognitive tests); structured learning input as digital training; Phase 2 in MCI/early AD and, separately, in neurodevelopmental populations (ANVISA and ethics-system scrutiny; assent plus proxy consent; independent data-safety monitoring).	Safety and biomarker signal.

Hard barriers that must be stated: no validated way exists to deliver FGF17 to the human brain; the mitogenic and angiogenic risk of an FGF-class CNS agent may never clear a paediatric risk-benefit assessment for a non-fatal indication such as a learning disability, and enhancement in people without disease carries a higher ethical bar than treatment of disease; 7,8-DHF is reported not to activate TrkB reliably in cell assays [2-4]; no human trials of FGF17 or 7,8-DHF were found (verify); healthy-animal enhancement may not transfer to disease or humans. The FGF23/phosphate effect is not the relevant class risk (a different, endocrine FGF).

What this study must deliver to earn Stage 1a: reagent activity and FGF17 pathway engagement (0a); a positive direction on the overall index (0b), with set-shifting reported separately; and, optionally, an EdU+/Olig2+ direction on banked tissue in the larger study. Covariation of structure with behaviour cannot be judged at this n.

3. Design and animals

Young adult male Wistar rats (about 8–10 weeks old, ≈ 250 – 300 g at surgery) from a CONCEA-registered animal facility (a university central biotério or institutional breeder; strain and sanitary status stated; SPF Wistar are bred in Brazil, for example at CEMIB/Unicamp, but no current price was found there, so supply, age-matched litters and price are confirmed directly with the facility). Males only. Housed on a reversed 12:12 light cycle so that dark-phase work falls in working hours, in pairs of the same arm and time block (single housing only with CEUA justification, identically in all arms). The sponsor randomises by baseline weight and source litter and holds the allocation and drug-code key (section 10). The 30 main rats are split into two halves (A and B, 5 per arm each) that take the tasks in different orders, and into five time blocks of six rats per day (two per arm, assigned at random within arm). Blinding: an unblinded host technician who does no testing or scoring prepares coded amber syringes and vials from the sponsor's key (7,8-DHF is a yellow flavone; a clear vehicle would reveal the arm); the injector and weigher never test or score; surgeon and behaviour operators are blind to drug. The environment arm cannot be fully blinded, so ELE handlers and test-day scorers are different people, set-shifting and Hebb-Williams are scored live by testers blind to environment where possible and verified from video under coded IDs, and 20% of videos are re-scored for inter-rater agreement (ICC). Arena identity is rotated by day within each arena type so that it is not aliased with arm.

Group	Content	Enrolled
Main cohort (one wave)	3 arms $\times$ 10: C (vehicle + standard), E (vehicle + ELE), D+E (FGF17 + 7,8-DHF + ELE); halves A/B of 5 per arm	30
Engagement satellite – FGF17	At 30 min after one ICV bolus ($0.5 \mu\text{g}$ in $3 \mu\text{l}$, as in the main cohort): time-matched vehicle 4, FGF17 4, heat-inactivated FGF17 (boiled aliquot, equal mass) 4; restraint duration logged; same tissue method in all groups; t_0 = injector removal; punch of the periventricular wall (right hemisphere) and hippocampus; pERK/total ERK	12
Practice/tolerability	2 cannula/dye placement checks; 4 vehicle-tolerability rats (2 low-DMSO solution, 2 CMC-Na suspension, each receiving 7,8-DHF in its vehicle for the full 35 days; weekly weights, daily abdominal-wall palpation score, necropsy scoring; a screen for gross intolerance only; switch rule: $>10\%$ weight loss or adhesion score $\geq 2 \rightarrow$ switch vehicle)	6
Total	Cannulated rats: $30 + 12 + 2 = 44$	48
Contingency (registered maximum, not enrolled unless needed)	One FGF17 re-test (4 vehicle + 4 FGF17 + 4 heat-inactivated, section 7) $\rightarrow$ registered maximum 60 rats, 56 cannulated	12

Enrolment is 10 per arm because ICV-cannula attrition in rats is commonly 30–40%; the plan assumes 62–69% completion (6–7 per arm). Cannula loss is logged by arm. Because climbing and wheels exist only in the enrichment arms, cap loss is expected to be higher there: protected recessed caps with a smooth collar, climbing structures ≥ 15 cm below the lid and solid-surface (rung-free) wheels are used, an arm-stratified tipping-point analysis is pre-specified, the delivered FGF17 dose is reported per D+E completer, and a completion difference between arms of >20 percentage points is flagged and >25 triggers a review (and possible stop). Welfare: the combined burden (cannula for ≈ 7 weeks, 35 daily injections, food restriction, swimming) is expected to be graded G13 (grau de invasividade) by the CEUA, with a scoresheet and humane endpoints: $>15\%$ weight loss from the growth-adjusted target, injection-site lesions, cannula infection, hypothermia rule, neurological signs after ICV (circling, head tilt, seizures: veterinary review and euthanasia if persistent), and a loosened or lost headcap (one re-capping attempt under brief anaesthesia; exposed skull $\rightarrow$ euthanasia). Post-operative analgesia uses locally available drugs; controlled drugs (ketamine, opioids) require the host's special ANVISA authorisation (section 9). Pre-specified exclusions: dye/histology-confirmed ventricular placement, humane endpoints, swim speed $<50\%$ of cohort median.

Power (honest). With 6–7 completers per arm, a pairwise comparison reaches 80% power only for $d \approx 1.6$ – 1.8 (unadjusted). Healthy young rats are near ceiling on simple tasks, and a cross-species enrichment meta-analysis reports $g \approx 0.44$ in rodents (an unpublished dissertation [15]), so a null result is likely even if a modest true effect exists, and only a very large effect would be seen. Pilot effect sizes must not be used to power the next experiment from the point estimate [12,13].

4. Interventions

Component	Specification
FGF17 (days 0–6)	Recombinant mouse FGF17, one 50 µg pack (mouse-rat sequence alignment documented before ordering). Chronic guide cannula to the right lateral ventricle (scaled coordinates). Four boluses on days 0, 2, 4, 6 in the D+E arm: 3 µl of ≈167 µg/ml (0.5 µg each; 2.0 µg per rat) in aCSF + 0.1% BSA (standard anti-adsorption practice, not a vendor specification) + heparin 10 µg/ml, awake, 0.5–1 µl/min with the injector left 2–3 min; vehicle boluses in C and E. The 167 µg/ml working stock lies within vendor reconstitution ranges (typically ≥100 µg/ml, up to 1 mg/ml; reconstituted FGF17 keeps 2–7 days at 4 °C), so aliquots are frozen at –80 °C in carrier and used within about 4 months, recovery after reconstitution and freeze-thaw and a pERK potency recheck (≈5 µg) are done before the main cohort, low-bind tubes are used, the solution is inspected for precipitate, and if unstable the concentration is lowered and the volume raised (up to 5 µl; the shortfall in delivered mass is then declared); dead volume is measured and primed so that 3 µl is delivered. Supply from the single pack: D+E 10 × 2.0 = 20 µg; satellites 4 × 0.5 + heat-inactivated 4 × 0.5 = 4 µg; contingency re-test 4 × 0.5 + 4 × 0.5 = 4 µg; bioassay ≈10 µg; potency recheck ≈5 µg; dead volume and repeats ≈5 µg; ≈48 µg of 50 µg, so there is no margin and a 25 µg vial is the contingency. <u>Delivery gap and dose anchor</u> : Iram 2022 infused continuously for about a week with an Alzet 1007D (≈84 µl delivered); one reviewer retrieved the Methods as recombinant mouse FGF17 at 25 µg/ml (≈2.1 µg continuous) plus a separate acute 0.5 µg bolus in 2 µl; another reviewer could not retrieve it, so the host must confirm it in the Methods (PMC9377328 or the UCL copy) before ordering. Four 0.5 µg boluses (2.0 µg total, 12 µl) approximately match that one reported total mass but not the exposure time: they are pulsatile and ≈7× smaller in volume, so a null remains confounded with delivery and FGF17 stays exploratory.
7,8-DHF (days 0–34)	5 mg/kg IP daily, days 0–34 (35 injections; rotated sites) in the D+E arm; vehicle IP in C and E. Daily order: testing (if scheduled) → 7,8-DHF injection → ELE/control session 30 min later, so the agent is on board during enrichment (mouse half-life ≈2 h) and the next day's testing falls at trough (≈18–24 h after dosing in mice; one rat report gives a terminal half-life of ≈23 h, route unclear — if that is right the drug accumulates and testing is not drug-free). Rat precedent for 5 mg/kg IP daily includes 5 weeks in a rotenone model in 9-month-old rats [17] and an optic-nerve model with outcomes at 7–21 days [18]; mouse work exists [5]. The 35-day duration is at the long end of that precedent. Vehicle (decided by the tolerability check before the main cohort is ordered) : default is a low-DMSO solution (5% DMSO / 40% PEG300 / 5% Tween-80 / 50% water per Selleck, or 10% / 40% / 5% / 45% saline per MedChemExpress/InvivoChem, ≈2.5 mg/ml; prepared fresh); fallback is an aqueous CMC-Na suspension (no peer-reviewed intraperitoneal PK found; an IP suspension may give erratic absorption, so a null could reflect exposure). Oil vehicles are avoided. Necropsy scoring of peritoneal adhesions/granulomas in all main-cohort rats; identical vehicle in all arms.
Enriched learning environment (ELE)	Days 0–34, one 60-min solo session per day (the test rat alone; no partner rats, which removes social contact as a confound and 12 partner animals) 30 min after the 7,8-DHF injection, in the dark phase under red light [6,7]. This is a weak enrichment dose and a non-social one (declared). Minimum specification the host must meet regardless of local habit: arena ≥1 m ² (the same size for control), ≥3 novel objects per session, running wheels (solid surface) and tunnels/climbing in each ELE arena, ≥3 puzzle devices on a six-level ladder (latch, string-pull, lever, chew), small fixed-calorie rewards (counted per rat; deducted from the ration only once food restriction has begun), engagement minutes and ladder level logged per rat. <u>Excluded from the ELE: odour discrimination and digging puzzles</u> . Control arena: same duration and handling, empty. SOP with photos completed by the host before CEUA submission.

5. Timeline and workload (day 0 = first bolus; one wave)

Days	Activity
–21 to –15	Arrival and 7-day acclimatisation; reversed light cycle.
–14 to –12	Cannula surgery with analgesia (≤ 10 rats/day); ≥ 10 days recovery before day 0.
–11 to –1	Recovery, handling, weighing; arena habituation from day –7 (arm-neutral).
0–6	FGF17/vehicle boluses (days 0, 2, 4, 6).
0–34	ELE/control sessions and 7,8-DHF/vehicle every day (35 doses), weekends included.
5–10	Gradual food restriction to 90–95% of the growth-adjusted target by day 10; held through the evening of day 28 (humane limit $< 85\%$); daily ration given 2 h after the end of each rat's own ELE/control session. Rewards in the ELE are counted per rat. Hours since feeding are recorded descriptively.
11–28	Tasks 1 and 2 run in parallel in two halves (A and B, 5 rats per arm each), in opposite orders. <u>Half A:</u> set-shifting days 11–22 (habituation 11–13, training to criterion 14–18, cap 5 days; tests start for 5 rats on each of days 19, 20 and 21 and last two consecutive days each: day 1 SD, CD, ID1; day 2 ID2, ED; last finish day 22), then Hebb-Williams days 23–28 (adaptation 23–24, then 4 problems on days 25–28). <u>Half B:</u> Hebb-Williams days 11–16 (adaptation 11–12, 4 problems on days 13–16), then set-shifting days 17–28 (habituation 17–19, training 20–24, tests start on days 25, 26, 27; last finish day 28). Set-shift tests have a 2.25-h cap per rat per day (a stage unfinished at the cap resumes next day; trials are capped at 60 per stage); the overnight gap between day 1 and day 2 is a declared deviation from single-session protocols; a one-session reminder (4 trials of simple discrimination) is given the day before a rat's first test. Peak: 10 rats in set-shift testing at once; ≥ 10 digging boxes are required. Food restriction ends on the evening of day 28.
29–34	Task 3 (last; non-appetitive): water maze for all 30 rats — acquisition days 29–32, probe day 33, visible-platform test day 34 (Appendix A).
35–37	Under anaesthesia an ICV dye injection (3 μ l) confirms patency, then decapitation (10 rats/day, balanced by arm and block, ≤ 12 min per animal); a 2-mm coronal slab through the cannula track is cut fresh from the right hemisphere and photographed; the rest of the right hemisphere is snap-frozen and banked at -80 °C; the left hemisphere is cut into 1.5–2 mm slabs, fixed 48 h, cryoprotected in sucrose and banked at -20 °C. Banked tissue is not analysed in this study.

Daily schedule and workload. Each day has five time blocks of six rats (two per arm). A rat's sequence is: test (if scheduled; ≤ 2.25 h) $\rightarrow$ 7,8-DHF/vehicle injection $\rightarrow$ ELE/control session 30 min later (60 min) $\rightarrow$ ration 2 h after the session ends. Block injections start at lights-off and every 75 min after it (so ELE sessions start 30 min after each injection). Rats scheduled for a test that day are placed in blocks 3–5 (injection at ≥ 2.5 h after lights-off, after the test has finished), which gives room for up to 18 testing rats a day; the last session ends ≈ 6.5 h and the last ration falls ≈ 8.5 h after lights-off. Six arenas are used (four ELE, two control), one operator per two arenas, so the working day spans ≈ 10.5 h (lights-off minus 1 h to $+9.5$ h) covered by two staggered shifts. Ordinary days need about 26 operator-hours (4–5 people); set-shift test days about 60 (up to ≈ 10 people, with ELE staff, the blinded injector and testers kept separate); the weekend rota repeats the same task schedule (day 0 should be a Monday so that surgery falls Monday–Wednesday). In-life: arrival day –21 to euthanasia day 37 = 58 days (8.3 weeks). Phase 0a (reagents, tolerability, satellites) comes first; its calendar is in section 9. The fixed task orders (A: set-shifting $\rightarrow$ Hebb-Williams; B: Hebb-Williams $\rightarrow$ set-shifting; water maze last for both) are declared and the half is a fixed effect in the analysis; ELE and DHF after testing form a post-training consolidation window, declared. Fallback if testers are scarce: set-shift tests run 2 rats a day; DHF and ELE then continue daily until each rat's last water-maze day (up to ≈ 45 doses, which the CEUA must approve in advance), with arms paired across consecutive days.

6. Endpoints and tissue

Primary analysis (per task). The three task scores are standardised with the grand mean and the pooled within-arm SD estimated in-sample (no prior data exist), winsorised on residuals at ± 3 , signs aligned. They are analysed in one linear mixed model (task as a repeated factor; random intercept per animal; time block as a random intercept if it does not collapse to the boundary; half (task order) as a fixed effect; arm $\times$ task terms; baseline weight as the only covariate; common residual variance, with task-specific variances if the fit allows), using all animals with ≥ 1 task, with Kenward-Roger intervals (SAS PROC MIXED with DDFM=KR is the primary software; R packages are used only as a sensitivity check, because KR with heterogeneous variances is not available there). The gate statistic is the index contrast (mean of the three z-scores) $D+E - C$ in per-task-SD units, with $D+E - E$ and $E - C$ reported; set-shifting alone and the index without the water maze are reported separately. Sensitivity analyses: complete cases; tipping-point (δ -adjusted) analysis for informative dropout (the mixed model is valid under missing-at-random only). Wheel running is zero in two arms and post-restriction weight is post-randomisation, so neither is a covariate. The inter-task correlation matrix is computed within arm from residuals and is descriptive. Analysis is run on coded labels by an analyst who does not hold the key (a third-party statistician if possible) until data lock.

Set-shifting score. Criterion 6 consecutive correct, maximum 60 trials per stage; the primary score is log(trials to criterion at the extradimensional shift, capped at 60), with rats that do not reach it assigned the cap. If more than 20% of rats hit the cap at ED, the rank/censored (Cox) analysis becomes primary instead (the OC table below assumes the uncensored log score); the validity check (ED > ID2 cohort-wide) always uses a censored comparison. Conditioning on the post-randomisation ID2 score is a sensitivity analysis only. Water maze: mean proximity to the platform in the 24-h probe [21] (start-position correction stated by the host); quadrant time also reported.

Block	Readouts	n
Behaviour	Three tasks; scored live and verified from coded video.	all completers
Banked tissue (every main-cohort animal)	Left-hemisphere slabs fixed and banked; right hemisphere frozen. No analysis in this study; reserved for EdU/Olig2 histology and synaptic markers in the larger study (immersion fixation gives lower and depth-dependent quality than perfusion — declared).	30
Engagement (satellites, snap-frozen)	pERK/total ERK in periventricular tissue (gate anchor) and hippocampus (descriptive); ice-cold dissection (<2 min), phosphatase inhibitors, FGF2-stimulated positive-control lysate and pre-specified assay CV on every blot; the same killing and dissection method in every group (decapitation without anaesthesia needs CEUA justification under the CONCEA euthanasia guideline; isoflurane would itself alter pERK); one gel carries all 12 samples plus the bridging control so fold-changes are compared on one membrane; pERK and total ERK on the same membrane (strip and re-probe, validated) or a parallel gel.	4 + 4 + 4

7. Statistics and pre-registration

Register on OSF before surgery (sponsor does this); ARRIVE 2.0 [14]; flow diagram of enrolled, excluded and completed animals. Estimation only: unadjusted contrasts with Kenward-Roger 95% intervals as primary; an animal-level (cluster) bootstrap that re-runs standardisation, winsorising and the model in every replicate is reported as sensitivity; no significance claims are made. The task order halves are balanced by arm, so the main contrast is protected, but arm × half cannot be assessed. Sizing of the next study: use the upper 80% confidence limit of the pilot's raw-unit residual SD per task [20] and the full factorial/interaction design.

8. Pre-registered decision rules and operating characteristics

Phase	Pass condition
0a – reagents	FGF17 bioassay active on the pack; mouse-rat alignment documented; DHF vehicle stable and tolerability screen passed (switch rule applied); potency recheck passed before the main cohort.
0a – FGF17 engagement (gate; signalling at the ventricular interface — necessary, not sufficient; the hippocampal punch is the descriptive parenchymal-penetration readout)	Periventricular pERK/total ERK, ratio of group means, FGF17 $\geq 1.5 \times$ time-matched vehicle AND FGF17 $\geq 1.25 \times$ heat-inactivated FGF17, at 30 min (n = 4 vehicle, 4 FGF17, 4 heat-inactivated; point estimates; acceptance CV pre-specified; positive-control lysate on every blot; the threshold applies to the diluted punch tissue actually assayed). A borderline result (FGF17/vehicle ratio 1.3 to <1.5) triggers one re-test (fresh 4 + 4 + 4 rats at 30 min, same pass rule; the decision rests on the fresh sample alone). Operating characteristics (200,000 simulations; lognormal assay noise at CV 30% / 50%; effect present; heat-inactivated group at vehicle level): false-pass with the re-test $\approx 1.6\%$ / 7.8% if there is no effect; pass $\approx 26\%$ / 30% at a true 1.3-fold change; $\approx 57\%$ / 49% at 1.5-fold; $\approx 97\%$ / 83% at 2-fold. A true 1.5-fold engagement therefore passes only about half the time: the gate detects large engagement and can miss a real but modest one. The single 30-min time point is an assumption anchored only by the FGF2 control curve and the FGF1 ICV precedent (peak at ≈ 20 min in hypothalamus); if the true peak is earlier or later the gate can miss it. FGF17, an FGF8-subfamily ligand, is expected to signal through c-isoform FGFRs (FGFR3c/FGFR4, per Zhang 2006 [23]; FGF17-specific profile to be verified); the cited oligodendrocyte-lineage FGFR study tested other FGFs, not FGF17 [16]. If the gate is not met, the main cohort does not start; test a higher dose or continuous infusion in a small follow-up. An FGFR-inhibitor arm is deferred to the follow-up experiment.
0b – feasibility	$\geq 50\%$ of the 30 enrolled rats (≥ 15) complete all three tasks; cannula placement $\geq 75\%$ correct. (Binomial $P(\geq 15/30)$: 0.77 at 55%, 0.90 at 60%, 0.97 at 65%, 0.99 at 70% true completion; a single wave, so no wave clustering.)
0b – direction	Index contrast D+E – C $\geq +0.5$ per-task SD (in-sample pooled SD; a small circularity that is declared) and D+E $\geq$ E in point estimate. The gate applies to the point estimate; the interval is reported alongside and is not itself a gate.
Provisional go	Phase 0a passed AND 0b feasibility AND 0b direction. Joint probability, illustratively assuming independence and 90% feasibility: $\approx 63\%$ if FGF17 truly engages at 2-fold and the target effect holds, $\approx 37\%$ if engagement is 1.5-fold; $\approx 10\%$ if FGF17 engages but behaviour is null; $\approx 0.2\%$ under a true global null (CV 30%; larger at CV 50%). Differential cap loss can select fitter D+E completers and bias the direction upward; the arm-stratified tipping-point analysis addresses this.

Operating characteristics of the direction rule (simulated; 6 completers per arm | 7 completers per arm; three tasks; index contrast; r = inter-task correlation; percentage that would be a “go” on direction alone):

True situation	r=0	r=0.3	r=0.6
No effect	7% 5%	11% 9%	13% 12%
Modest (E 0.3, D+E 0.5)	45% 45%	43% 43%	41% 42%
Target (E 0.3, D+E 0.8)	79% 81%	72% 74%	67% 69%
Large (E 0.5, D+E 1.2)	97% 98%	92% 94%	87% 89%
Enrichment alone (E 0.6, D+E 0.6)	39% 40%	39% 38%	38% 38%

The rule is lenient (about 10% false-go) and cannot tell the drug package from enrichment: enrichment alone gives a “go” in about 39% of runs. It only screens for a promising direction. A “go” justifies the larger study only; a “no-go” identifies the failure mode (delivery, engagement, ceiling, attrition) and is not evidence that the approach cannot work.

9. Brazil logistics and budget

Item	What the review found
Ethics	Animal use is approved by the host institution’s CEUA (Comissão de Ética no Uso de Animais) under Lei 11.794/2008 (Lei Arouca), Decreto 6.899/2009 and the CONCEA normative resolutions. The institution, its CEUA and its animal facilities are registered in CIUCA, and the institution must hold a valid CIAEP (CONCEA accreditation); check both on the CONCEA list before choosing a host, because a lapsed accreditation bars animal work. The CEUA form grades invasiveness (GI1–GI4; this study is expected to be GI3). Every person on the protocol needs documented training (CONCEA RN 49/2021). Review time differs by institution (1–6 months; a reviewer assumption); ask for an interim review. The principal investigator must be host staff. Animals are reserved only after approval; the practice and contingency rats are in the approved protocol.
Imports	FGF17 and antibodies come in one shipment (a single 50 µg pack; later orders only if the contingency is used); about 4–12 weeks including licensing, customs and cold chain (reviewer assumption), customs-broker (despachante) fees, dry-ice courier. Duty relief: Lei 8.010/90 (II, IPI), with PIS/COFINS-import relief under Lei 10.865/2004 as applied in CNPq Portaria 2.248/2025, for imports by CNPq-accredited institutions or researchers who are the importer of record; the licence is registered in Siscomex and needs CNPq approval (legal review up to 60 days) — confirm with the host’s import office whether the licence must precede payment. ICMS relief (Convênio ICMS 93/98, adopted state by state) applies only if the import is also exempt from II/IPI, so losing the CNPq exemption also loses the ICMS relief. Reagents under sanitary control follow ANVISA RDC 172/2017 (automatic approval for CNPq-accredited importers, except human biological samples and substances controlled under Portaria SVS/MS 344/98). Importa Fácil Ciência (Correios; postal origin only, ≤30 kg, historically up to USD 10k) is probably unsuitable for dry-ice cold chain. Risk: the annual CNPq quota is regularly used up (2025: USD 229M, frozen mid-year; 2026: USD 254M, exhausted in June and raised to ≈USD 354M in July), after which licences are refused or deferred to the next year; confirm the quota position with CNPq and the host before ordering. If the exemption is refused, imported lines (≈2.9k landed) cost more, a risk scenario estimated at +1.0–1.5k (unverified) that is shown separately below. The base budget assumes landed cost ≈1.3–1.5× list (exemption working plus broker fees and state charges). Rates here come from search snippets and reviewer knowledge, not verified.
Controlled drugs	Ketamine and opioid analgesics are controlled (Portaria SVS/MS 344/98): the host needs ANVISA’s special authorisation (AEP for academic institutions) and a registered veterinary lead (CRMV); plan non-controlled alternatives (meloxicam, tramadol or dipyrone for analgesia) where permitted.
Equipment access (precondition)	The budget assumes written access to: stereotaxic frame, water maze + tracking software, set-shift and maze apparatus, cryostat, western blot system and image-analysis software. Multi-user facilities in Brazil often charge hourly or require a contrapartida; those fees are NOT in any scenario here and must be quoted.
Personnel, services, overhead	Not in the budget. About 1,900 operator-hours (section 5 and Appendix B), a surgeon, a coordinator and a veterinary lead for about 3–4 months including Phase 0a. If the operators are graduate students or technicians on stipends (unverified rates of roughly R\$17–30 per hour, i.e. about USD 3–5.5), the operator cost is on the order of USD 6–10k; with the other staff and institutional overhead (institution-specific; reviewer examples 15–35%) the plausible all-in total is USD 15–30k (an unverified order of magnitude). Student stipend rules (CAPES/CNPq/FAPs), dedication limits, night and weekend access to the animal facility and thesis/authorship expectations must be agreed with the host.
Timeline	Contracting and the work plan (2–6 months, not counted); CEUA 1–6 months; FGF17 procurement with one import (4–12 weeks) → bioassay (≈2 weeks); Phase 0a 2–3 months; in-life 8.3 weeks; analysis 1 month. About 6–11 months from CEUA submission to core results. Institutional risks (strikes, closures) are a host-selection criterion.

Phase 0a calendar (planning offsets in weeks before the main cohort arrives on day –21; the host confirms): CEUA approval and FGF17 order (–24); pack arrives, bioassay (–14 to –12); tolerability rats start dosing (–14) and finish at day 35 (–9); satellite and practice surgeries (–12 to –11) with dosing and tissue at –10 to –9; westerns and gate readout (–8 to –6); written decision (–5); main-cohort rats ordered (–4, after the gate and vehicle decision). Satellite and pilot surgery never shares a day with main-cohort surgery.

Budget (USD; list prices from search plus reviewer knowledge; to be replaced by quotes):

Line (core = Phase 0a + 0b)	Base	Notes
Rats: 48 (30 main, 12 satellites, 6 practice/tolerability) at external-rate prices, with transport	0.8k	Public price lists found: USP Wistar R\$62.50 external (2022), UFMG Biotério Central 2026 about R\$58–95 by age (strain column unconfirmed); internal rates about 0.35k lower

Line (core = Phase 0a + 0b)		Base	Notes
Housing/care ($\approx 2,250$ rat-days $\approx 1,100$ pair-housed cage-days, reversed-cycle room)		1.0k	Brazilian per-diem not found (ask the facility); India anchor USD 0.47–0.94 per cage-day; optimistic 0.55k, pessimistic 1.4k
FGF17, one 50 μ g pack, landed		0.65k	Cohesion $\approx$ USD 480/50 μ g via an aggregator; R&D; 25 μ g mouse vial price not found; PeproTech lists human only; a 25 μ g contingency vial 0.45k in pessimistic
7,8-DHF $\approx$ 2 g and vehicle components, landed		0.25k	TCI $\approx$ USD 62/g
Cannulas, dummies, screws, cement (44 rats)		0.3k	
Anaesthesia, analgesia, surgical and terminal consumables (44 cannulations, 48 terminal procedures), including controlled-drug handling		0.5k	
ELE: 6 arenas (4 ELE, 2 control), solid wheels, objects, puzzle devices		0.5k	Get two local quotes; ICMS/IPI are embedded in the invoice price
Behaviour consumables and fabrication (≥ 10 digging boxes, set-shift pots/media/odours, 2 Hebb-Williams mazes)		0.6k	
ERK block, landed: pERK, total ERK, HRP secondaries, loading control, FGF2 positive control		1.3k	list $\approx$ 1.15k
Western reagents (membranes, ECL, inhibitors)		0.35k	
Import shipment with customs-broker and courier fees		0.4k	Brazilian broker and storage fees not quoted
CEUA fee and miscellaneous		0.1k	
Contingency rats (12 registered re-test rats: purchase, housing, surgery)		0.5k	Spent only if the re-test is triggered
Core base total, consumables only		7.3k	excludes personnel, overhead, equipment fees
Scenario	Core	Assumptions (itemised)	
Optimistic	$\approx$ 6.0k	host animal facility at internal rates (rats -0.35 k, housing -0.45 k), antibodies at list -0.4 k, cheaper shipment -0.1 k	
Base	$\approx$ 7.3k	as above	
Pessimistic	$\approx$ 8.8k	15% overrun plus a contingency 25 μ g FGF17 vial; equipment fees, personnel and overhead still excluded	
Pessimistic plus refused CNPq exemption	$\approx$ 10.3k	imported lines taxed ($+1.0$ – 1.5 k, unverified); also likely months of delay	

Ceiling (researcher-approved): USD 12k for materials. All scenarios, including the tax-paid import case, are inside it. Phases: 0a $\approx$ 4.4k (satellite, practice and contingency rats, the FGF17 pack, ERK antibodies, surgery, shipment) with a stop rule at that spend; 0b $\approx$ 2.9k. First cuts if prices run high: enrol 8 per arm (the feasibility gate becomes ≥ 12); use the host's internal rates. The western block must not be cut (it carries the only gate). Must not be cut: dye/histology placement check, FGF17 bioassay and alignment, blinding/randomisation, DHF tolerability check, visible-platform control, banking of tissue.

10. Lab handover and remote oversight

The researcher will not necessarily be on site; the host laboratory runs the experiment. This PDF is the working protocol; the following must exist before animals arrive. SOPs, CEUA forms and scoresheets are kept bilingual (Portuguese for staff; English for the sponsor).

Item	Responsibility / requirement
Legal form (decide first)	Either (a) an academic partnership with the host institution (an acordo de parceria para PD&I; under Lei 10.973/2004, or a service contract under art. 8), usually with funds received and managed by its research-support foundation (fundação de apoio, Lei 8.958/94) as intervening party; the overhead is set by the institution; IP and data ownership are written into the instrument and the host's technology-transfer office (NIT) is usually involved; or (b) a fee-for-service contract (prestação de serviços) with an invoice (nota fiscal), ISS and taxes agreed. The sponsor's residence (in Brazil or abroad) decides currency exchange, IOF and withholding and must be stated; whether goods imported duty-free under Lei 8.010 may be used in a paid service contract must be confirmed with CNPq or the import office. Foreign funds arrive by international transfer with the usual exchange documentation; no restriction like India's FCRA is known to us, but confirm in writing with the foundation. The choice determines payment mechanics and is a precondition for CEUA submission.
Host selection and due diligence	Criteria: institution with a valid CONCEA CIAEP, a functioning CEUA and a registered animal facility; ANVISA special authorisation for controlled drugs and a registered veterinary lead (CRMV); a dedicated reversed-cycle room, six arenas, a water maze, two Hebb-Williams mazes and a separate euthanasia room; written record of ICV cannulation, set-shifting and water-maze work; access to all listed equipment; references from two previous external collaborations; a named roster of operators (with training records) and a facility-hours map for nights and weekends; stable operation (no recent strike disruption). Trade-off: a CRO offers fee-for-service and quality systems but rarely these assays; an academic lab offers expertise but expects authorship and slower timelines.

Item	Responsibility / requirement
Payment tied to gates	Public hosts and foundations generally spend only deposited funds, so the first tranche must pre-fund Phase 0a: (1) at signature (CEUA fee, practice and satellite rats, the FGF17 pack and ERK antibodies, surgery, personnel for 0a); (2) after the host-competence gate and the 0a gate decision, the main cohort; (3) at data lock, a small holdback. A failed 0a stops spending at its tranche; staff commitments on monthly stipends are stated for that case. The agreement states who bears currency risk: budget in BRL, convert per tranche, and hold a 10–15% FX reserve (or agree USD tranches with a renegotiation band); bank spread and IOF are included.
Governance and welfare oversight	Protocol amendments need the host PI, the sponsor and the CEUA; the veterinary lead can stop the study; weekly remote welfare report (weights, scores, cap losses, adverse events); escalation: sponsor within 48 h, CEUA coordinator and unit head within 7 days; fixed weekly call slot.
Quality standard	Non-GLP; the host works to GLP-like good practice (contemporaneous records, calibrated equipment, training records, retained raw data).
Roles and accountability	Host PI (accountable for CEUA compliance and animal welfare; also the CEUA PI); study coordinator (one named person); about five to ten trained operators in staggered shifts as in section 5 (ELE, blinded injector, testers, preparer kept separate), a surgeon and a weekend rota; an unblinded technician outside the testing team who prepares coded syringes; sponsor (design, randomisation, drug-code key, registration, independent analysis).
Agreement terms	Funding, overhead and payment schedule; who pays for overruns and for a failed Phase 0a; IP ownership; data and tissue ownership; insurance and liability; stop rules; publication and authorship (ICMJE criteria); data sharing.
Host-competence gate (before Phase 0a)	Practice-group evidence: $\geq 80\%$ successful ICV cannulation by dye (if the gate fails, the sponsor funds a short training and re-test or reassigns the host, and no main-cohort money is spent until it is met); western blot CV from the lab's previous work; set-shifting and water-maze records; verified equipment list. Sponsor decides go/no-go on this evidence.
SOP pack (host completes with photos; sponsor reviews)	Surgery and cannula care; ICV bolus injection; vehicle preparation and dosing; ELE and control sessions; set-shifting; Hebb-Williams; water maze incl. temperature/hypothermia rules; food restriction and weighing; euthanasia, tissue banking; satellite dissection and western protocol; per-day time-budget table; welfare scoresheet; data-recording SOP; Appendix A parameters signed off. The CEUA application is drafted jointly: the sponsor supplies a model narrative (severity, endpoints, justification), the host adapts it.
Blinding and randomisation	Sponsor generates the allocation and the coded vial/arm labels (an unblinded host technician outside the testing team prints and fills them), and keeps the key; host testing staff handle coded IDs only; unblinding after data lock.
Data integrity and monitoring	Bound or electronic lab notebooks with audit trail; raw files hashed on upload; protocol deviations classed minor/major with major ones reported within 48 h and a sponsor decision; scheduled live-video audits and source-data verification of $\approx 10\%$ of records; weekly summary (enrolment, weights, exclusions, adverse events, cap losses); raw data and images shared with the sponsor for independent blinded analysis.
Tissue transfer	Banked tissue stays at the host unless a material transfer agreement and the necessary export permits are in place (export permit requirements to be confirmed).
Decisions	After 0a and 0b the sponsor issues a written decision (provisional go / stop / redesign) before further spending.

Appendix A. Task parameters (defaults; host to confirm and sign off before animals arrive)

Task	Parameters
Set-shifting (digging; Birrell & Brown 2000; Tait et al. 2018 [11,19])	Two ceramic pots per trial; stimulus dimensions: digging medium and odour (neither is used in the ELE); 3 days digging habituation to a baited pot; example exemplar sets (host to fix ≥ 6 each before animals arrive): odours — cinnamon, nutmeg, clove, cumin, thyme, vanilla; media — sawdust, shredded paper, sand, plastic beads, wood chips, cotton; stages: simple discrimination, compound discrimination, first intradimensional shift (ID1), second intradimensional shift (ID2, to build the attentional set), extradimensional shift (reversal stages omitted to limit session length — declared); validity check: ED trials > ID2 trials cohort-wide; dimension order and exemplar sets counterbalanced across rats and arms; odour applied to the medium and bait buried in both pots (masked), with a fixed discovery-trial rule; criterion 6 consecutive correct; maximum 60 trials per stage; 3-min trial limit; test cap 2.25 h per rat per day; error on first incorrect dig; bait $\approx \frac{1}{2}$ cereal piece; pot positions pseudo-random; training cap 5 days.
Hebb-Williams (Rabinovitch & Rosvold 1951 [22])	Closed-field maze with barrier layouts; 2 days adaptation with food reward including the standard practice problems; 4 test problems chosen from the standard 12 by host and sponsor before animals arrive (fixed order, same layouts for all rats; investigator-chosen subset, not validated); 6 trials per problem (3-min cutoff), one problem per day (4 days), two identical mazes in parallel; error zones defined in advance; split-half reliability reported.
Morris water maze	Pool ≈ 150 –180 cm, 25 ± 1 °C, hidden 10-cm platform 1.5–2 cm below the surface in opaque water (non-toxic black tempera paint, for contrast with albino rats), one pool, ≥ 4 distal cues, dim white light (≈ 10 –20 lux, measured; spectrum recorded) in the pool room only — a declared exception to the dark-phase red-light rule; rats travel in covered cages and the colony room stays dark; 4 trials/day, 60 s maximum, four quasi-random start points, ≥ 5 min between trials in a warm cage (32 °C); core temperature before and after sessions (rest day if <34.5 °C); 4 acquisition days; 60-s probe 24 h after the last acquisition day; visible-platform test (4 trials) after the probe; scores: mean proximity (Gallagher), target-quadrant time, swim speed, floating, thigmotaxis.
ICV cannula (to be confirmed in the practice group)	Right lateral ventricle, rat aged 8–10 weeks (≈ 250 –300 g): from bregma AP -0.9 mm, ML $+1.5$ mm, DV such that the injector tip lies at -4.0 to -4.5 mm (the host states whether DV is measured from bregma or the skull surface; guide ≈ -3.0 to -3.5 mm with the injector protruding ≈ 1 mm), scaled by the bregma-lambda distance and confirmed by dye in the practice group; fixed with screws and dental cement under a protected recessed cap; dummy cannula between injections.
FGF17 bioassay (0a reagent gate)	Serum-starved FGF-responsive cells (the supplier's BALB/3T3 line, or NIH-3T3; primary rat OPCs if available) stimulated for 10–15 min with 0–100 ng/ml of the supplied FGF17 in the presence of heparin 10 μ g/ml (an EC50 comparison is only valid with matching heparin); pERK/total ERK by western; FGF2 as positive control; pass = a clear dose-response with ≥ 3 -fold maximal pERK over unstimulated cells; the supplier's stated ED50 is a proliferation value, so on first purchase the pERK dose-response is judged against FGF2 and the ≥ 3 -fold criterion (an earlier-lot comparison is only possible for later lots). A 3T3 pass confirms potency through FGFR1c/2c, not OPC-relevant FGFR3/4 signalling.
Antibodies and kits (host to confirm; unverified examples from memory)	pERK1/2 Thr202/Tyr204 (e.g. CST #4370), total ERK1/2 (e.g. CST #4695); the host lists final catalogue numbers, rat reactivity and dilutions in the SOP pack.
ELE / control session	Dark phase under red light; solo session of 60 min (no partner); 6 arenas (4 ELE, 2 control), arena identity rotated by day within type; arms balanced within every time block (two rats per arm per block); log object list, puzzle level, rewards, engagement minutes, wheel revolutions.

Appendix B. Operator time budget (to be confirmed by the host)

Period	Daily work	Operator-hours/day
Days –14 to –12	Cannula surgery, ≤10 rats/day (surgeon + assistant)	≈ 24
Days 0–34, every day incl. weekends (routine)	ELE/control: 30 solo sessions in 6 arenas over five blocks (≈6.5 h of sessions; one operator per two arenas ≈ 20 h); injections ≈ 1.5 h; weighing, ration and cage checks ≈ 1.5 h; cleaning and object preparation ≈ 2 h; logs and video ≈ 1 h	≈ 26
Days 0, 2, 4, 6 (extra)	ICV boluses (30 rats × ≈10 min × 2 operators)	+ 10
Days 5–10	Food restriction ramp, extra weighing	+ 2
Days 11–16	Half A set-shift habituation/training (15 rats × ≈1 h); half B Hebb-Williams (15 rats × ≈0.5 h)	+ 22 (total ≈ 48)
Days 17–18	Half A training 15 h; half B set-shift habituation 15 h	+ 30 (total ≈ 56)
Days 19–22	Half A set-shift tests: up to 10 rats testing × ≤2.25 h ≈ 22 h; half B training ≈ 15 h	+ 37 (total ≈ 63)
Days 23–28	Half A Hebb-Williams ≈ 8 h; half B set-shift training then tests ≈ 22 h	+ 30 (total ≈ 56)
Days 29–34 (water maze)	30 rats × 4 trials, ≥5 min between trials, two core temperatures ≈ 9 h with two operators	+ 18 (total ≈ 44)
Days 35–37 (euthanasia)	10 rats/day × ≤12 min with two operators plus tissue handling; no routine (ELE and dosing end on day 34)	≈ 8
Weekends	The same task schedule repeats on weekends (day 0 on a Monday)	as the weekday row

Total ≈ 1,900 operator-hours for the main cohort and Phase 0a. The peak is ≈ 63 operator-hours a day on set-shift test days (days 19–22), which needs up to about 10 people in staggered shifts (ELE staff, the blinded injector and the testers are different people); ordinary days need 4–5. Personnel cost is not in the budget.

11. Validation record

The mini design inherits its task methods, assay logic and many safeguards from a larger design reviewed in the rounds below (the early rounds were run in the India form). Reviewer panels were 5 Opus + 2 Sonnet agents per round unless stated. Reviewer statements are model judgements, not independent human review.

Round	Reviewers	Verdicts	Main issues → fix
Earlier A, B, translational	3 Opus, 3 Opus + 1 Haiku, 1 Opus	All approve with changes	Pump ends before testing, pump cost, middle-aged rats, transfer overlap, social confound, composite not a g factor, overclaiming, neurotrophin history, FGF safety → boluses, young adults, staged gates, hedged roadmap.
India-form rounds 1–8	5 Opus + 2 Sonnet each	All approve with changes every round	Gate calibration, slab fixation, vehicle choice, sequential waves, fixed slots, set-shift scoring, staffing arithmetic, frozen-SD rule, Cox primary, time-matched vehicle, heat-inactivated control, tolerability, budget chain, many wording fixes (history of the larger design).
Brazil full-size round B1	5 Opus + 2 Sonnet	All approve with changes; budget and executability: major revision	Housing and rats underpriced; all-in cost unquantified; wrong legal instruments; CNPq quota exhausted in 2026; controlled-drug licences; dark-phase arithmetic on test days; weekend contradiction; staffing below concurrency; ill-defined gate and re-test → this mini redesign (smaller, one wave, solo ELE sessions, parallel halves, gate and OC re-simulated, budget re-itemised).
Mini round M1	5 Opus + 2 Sonnet	(pending)	
Mini round M2	5 Opus + 2 Sonnet	(pending)	
Mini round M3	5 Opus + 2 Sonnet	(pending)	

12. Items the reviewers still do not like / remaining gaps

#	Gap	Kind
1	Bolus FGF17 is not the Iram infusion ($\approx 7\times$ less volume, pulsatile); the dose approximately matches Iram's reported $\approx 2.1 \mu\text{g}$ total mass, but that figure rests on one reviewer's retrieval and exposure time is not matched; a null remains confounded with delivery.	Pumps in the larger study
2	Power: only very large effects detectable ($d \approx 1.6\text{--}1.8$ at 6–7 per arm); healthy rats near ceiling; the direction rule is lenient ($\approx 10\%$ false-go; $\approx 39\%$ go for enrichment alone) and cannot separate the drug from the environment because there is no drug-only arm.	Intrinsic to a mini pilot
3	The FGF17 gate uses one time point (30 min) and 4 rats per group; a true 1.5-fold engagement passes only about half the time and the peak time is an assumption; periventricular pERK shows bioactivity at the ventricular interface only.	Intrinsic; assay needed
4	7,8-DHF may be inert in vivo; 35 injections at the long end of precedent; the tolerability screen (2 rats per vehicle) only catches gross intolerance; DHF is not gated.	Declared
5	The ELE is solo, non-social and a weak dose; set-shifting is the only clean transfer readout; the arm cannot be fully blinded; parallel halves use opposite task orders (order effects are a fixed effect, arm $\times$ half not estimable).	Declared
6	Prices unverified (Brazilian per-diem not found); personnel, overhead and equipment fees excluded; all-in cost probably USD 15–30k (unverified); the CNPq import exemption and quota are uncertain (a refusal adds $\approx 1.0\text{--}1.5\text{k}$ and months).	Quotes and written confirmations
7	Host executability depends on a named roster (up to ≈ 10 people on busy days), rooms and equipment, CEUA/CIAEP status, ANVISA controlled-drug authorisation, a legal instrument and a pre-funded first tranche, none of which are in hand.	Host to supply
8	Structure (EdU/Olig2) and synaptic readouts are not analysed in this study; tissue is only banked.	Larger study
9	Optogenetics (Stage 1b) and the human roadmap are aspirations without design detail or validated tools in the host lab.	Own review round
10	Literature checks used search snippets only; Iram's delivered dose was reported by one reviewer only; the FGF1 ICV pERK precedent is not referenced; human-trial and neurotrophin statements to be verified; Brazilian legal facts are from snippets and must be re-checked by the host.	Researcher/host to verify

13. References and verification status

Checks were by search snippets only (PubMed, PMC, Crossref and publisher pages unreachable). “Verified” means consistent across indexed records.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. <i>Nature</i> 2022;605:509–515. doi:10.1038/s41586-022-04722-0 (an Author Correction exists)	Verified; delivered dose (25 $\mu\text{g}/\text{ml}$; 0.5 $\mu\text{l}/\text{h} \times 7 \text{ d} \approx 2.1 \mu\text{g}$) reported by one reviewer, not independently confirmed
2	Boltaev U, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. <i>Sci Signal</i> 2017;10(493). doi:10.1126/scisignal.aal1670	Verified
3	Todd D, Gowers I, Dowler SJ, et al. <i>PLoS One</i> 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; DHF content via reviews
4	Pankiewicz P, et al. Do small molecules activate the TrkB receptor in the same manner as BDNF? Limitations of published TrkB low molecular agonists and screening for novel TrkB orthosteric agonists. <i>Pharmaceuticals</i> 2021;14(8):704. doi:10.3390/ph14080704	Verified (review)
5	Rawlings-Mortimer F, et al. 7,8-dihydroxyflavone enhances long-term spatial memory and alters brain volume in wildtype mice. <i>Front Syst Neurosci</i> 2023;17:1134594. doi:10.3389/fnsys.2023.1134594	Verified
6	van Praag H, Kempermann G, Gage FH. <i>Nat Rev Neurosci</i> 2000;1:191–198. doi:10.1038/35044558	Verified
7	Nithianantharajah J, Hannan AJ. <i>Nat Rev Neurosci</i> 2006;7:697–709. doi:10.1038/nrn1970	Verified
8	Galsworthy MJ, et al. <i>Behav Genet</i> 2005;35:675–692. doi:10.1007/s10519-005-3423-9	Verified
9	Matzel LD, Han YR, Grossman H, et al. <i>J Neurosci</i> 2003;23(16):6423–6433 (general learning ability in mice). doi:10.1523/JNEUROSCI.23-16-06423.2003	Verified
10	Gibson EM, Purger D, Mount CW, et al. <i>Science</i> 2014;344:1252304 (mouse optogenetic premotor-cortex stimulation promotes oligodendrogenesis and myelination)	Verified
11	Birrell JM, Brown VJ. Medial frontal cortex mediates perceptual attentional set shifting in the rat. <i>J Neurosci</i> 2000;20(11):4320–4324 (published protocols differ in stages; some add ID2, which is used here as a modification)	Verified

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12	Kraemer HC, Mintz J, Noda A, Tinklenberg J, Yesavage JA. Caution regarding the use of pilot studies to guide power calculations for study proposals. Arch Gen Psychiatry 2006;63:484-489. doi:10.1001/archpsyc.63.5.484	Verified
13	Leon AC, Davis LL, Kraemer HC. The role and interpretation of pilot studies in clinical research. J Psychiatr Res 2011;45:626-629. doi:10.1016/j.jpsychires.2010.10.008	Verified
14	Percie du Sert N, et al. PLoS Biol 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified
15	Enrichment meta-analysis (rodent cognition $g \approx 0.44$, 95% CI 0.25-0.62): Fitzgerald, Univ. at Albany dissertation, a cross-species meta-analysis of environmental enrichment effects on cognition (scholarsarchive.library.albany.edu/etd/371); Léger M, et al. Environmental enrichment duration differentially affects behavior and neuroplasticity in adult mice. Cereb Cortex 2015;25(11):4048-4061. doi:10.1093/cercor/bhu119	Dissertation is not peer-reviewed; Léger verified
16	Fortin D, Rom E, Sun H, Yayon A, Bansal R. Distinct FGF/FGF receptor signaling pairs initiate diverse cellular responses in the oligodendrocyte lineage. J Neurosci 2005;25:7470-7479. doi:10.1523/JNEUROSCI.2120-05.2005	Verified
17	Nie S, et al. 7,8-DHF protects nigrostriatal dopaminergic neurons from rotenone-induced neurotoxicity in rodents. Rotenone model in 9-month-old rats, 7,8-DHF 5 mg/kg IP daily for 5 weeks. Parkinsons Dis 2019;2019:9193534. doi:10.1155/2019/9193534 (PMC6421741)	Verified
18	Gallego-Ortega A, et al. 7,8-Dihydroxiflavone maintains retinal functionality and protects various types of RGCs in adult rats with optic nerve transection. Int J Mol Sci 2021;22(21):11815. doi:10.3390/ijms222111815 (5 mg/kg IP daily; outcomes at 7, 14 and 21 days; the 1% DMSO detail was not confirmed)	Verified
19	Tait DS, Bowman EM, Neuwirth LS, Brown VJ. Assessment of intradimensional/extradimensional attentional set-shifting in rats. Neurosci Biobehav Rev 2018;89:72-84	Verified
20	Browne RH. On the use of a pilot sample for sample size determination. Stat Med 1995;14:1933-1940	Verified (title to check)
21	Gallagher M, Burwell R, Burchinal M. Severity of spatial learning impairment in aging: development of a learning index for performance in the Morris water maze. Behav Neurosci 1993;107:618-626. doi:10.1037/0735-7044.107.4.618	Verified
22	Rabinovitch MS, Rosvold HE. A closed field intelligence test for rats. Can J Psychol 1951;5:122-128 (Hebb-Williams maze)	Verified (issue number unconfirmed)
23	Zhang X, et al. Receptor specificity of the fibroblast growth factor family: the complete mammalian FGF family. J Biol Chem 2006;281:15694-15700 (FGF8 subfamily expected to act through FGFR3c/FGFR4; FGF17-specific profile not retrieved)	Real; FGF17 result unverified

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young rodents; any study combining 7,8-DHF with enrichment; rat ICV FGF17 doses; independent confirmation of Iram 2022's delivered dose; Brazilian prices for Alzet pumps and recombinant FGF17; a peer-reviewed validation of a Hebb-Williams subset; a Brazilian per-diem housing cost. Brazilian regulatory and price sources checked by search (snippets only; to be re-checked by the host): Lei 11.794/2008 and Decreto 6.899/2009; CONCEA CIUCA manual and RN 49/2021; CNPq Portaria 2.248/2025; Unicamp Comunicado SEC PRP 001/2025; FUNPEC 2025 and FUJB 2026 quota notices; ANVISA RDC 172/2017; Convênio ICMS 93/98; Lei 10.973/2004, Lei 8.958/94; USP Rede de Biotérios and UFMG Biotério Central price lists.